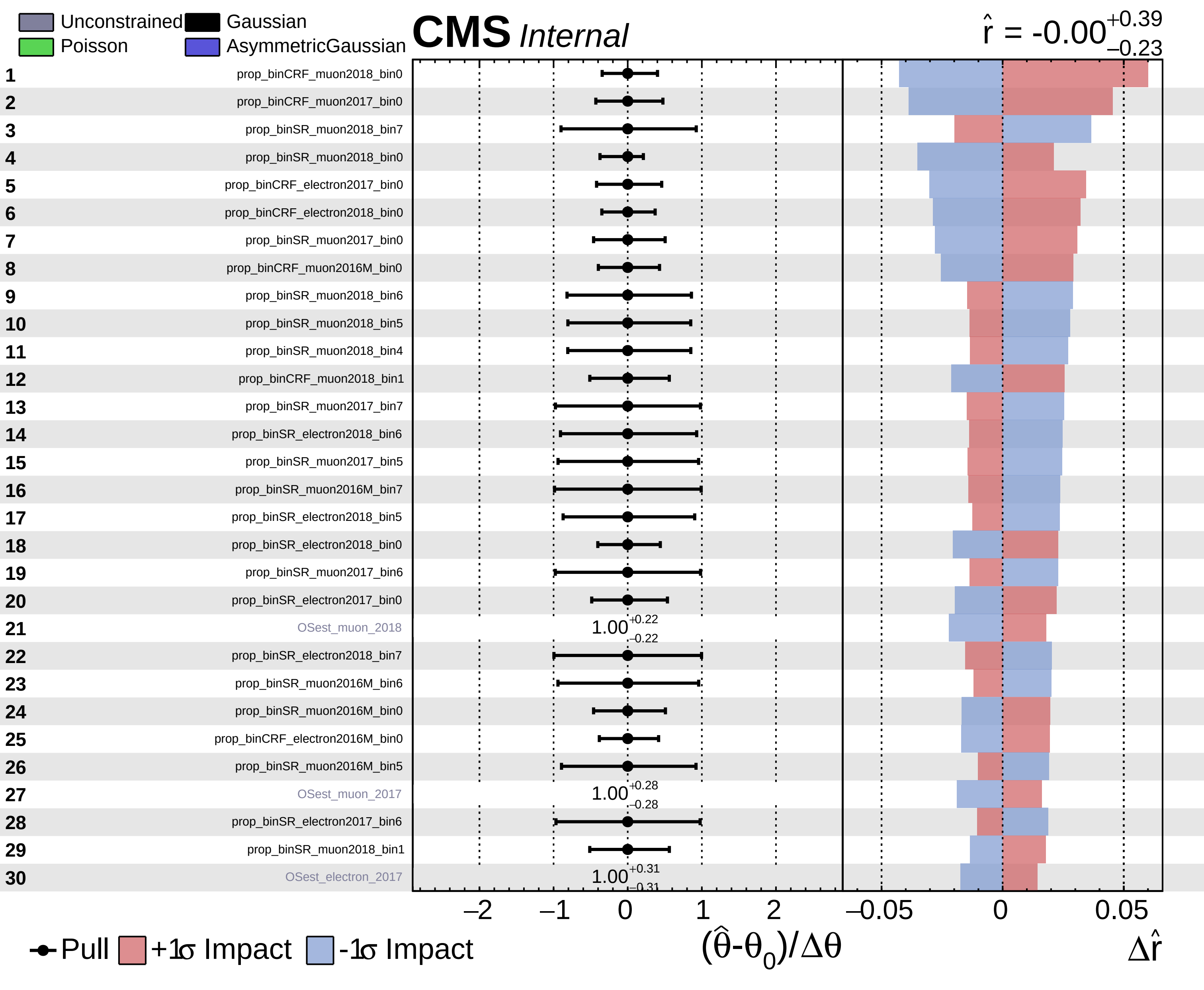
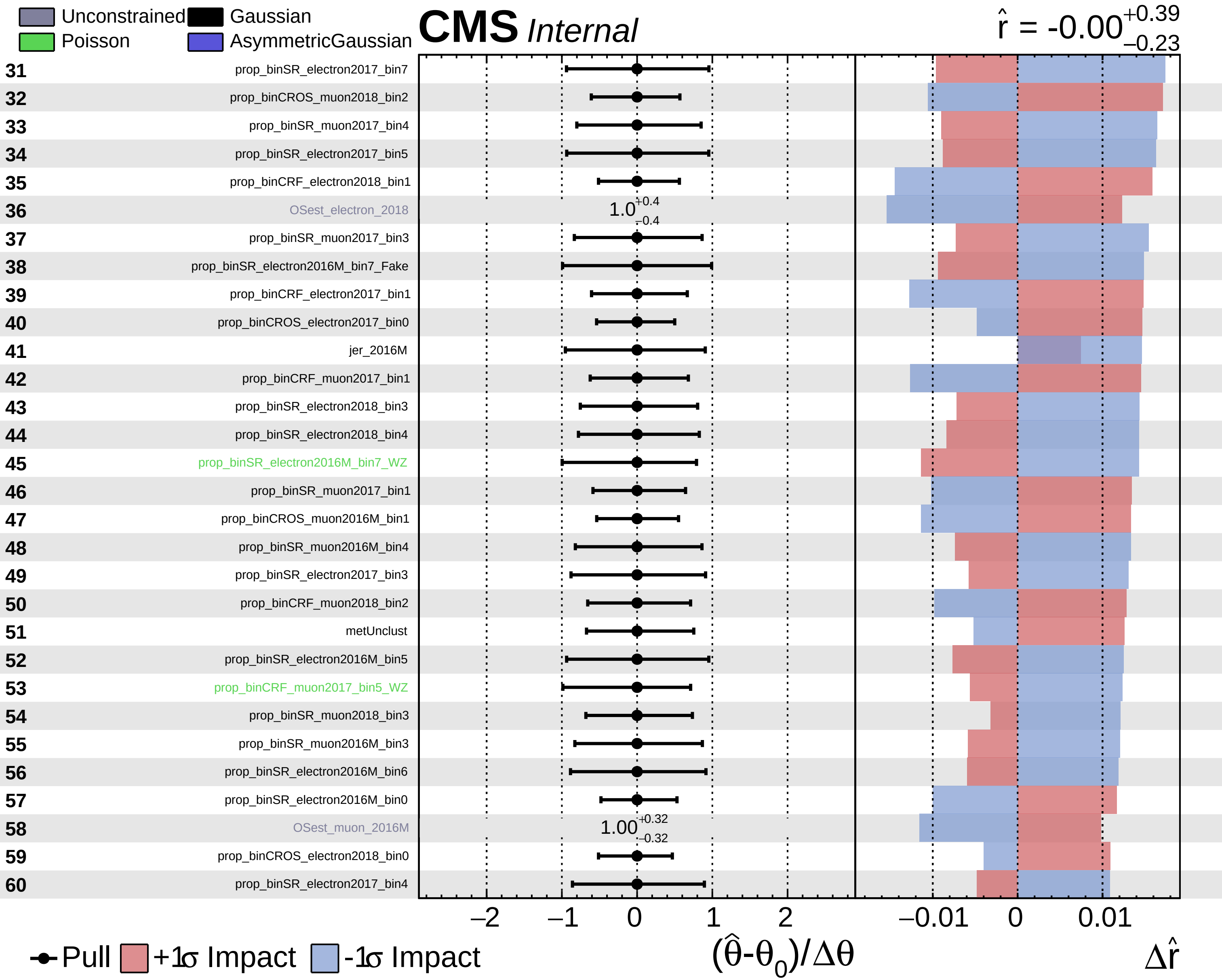
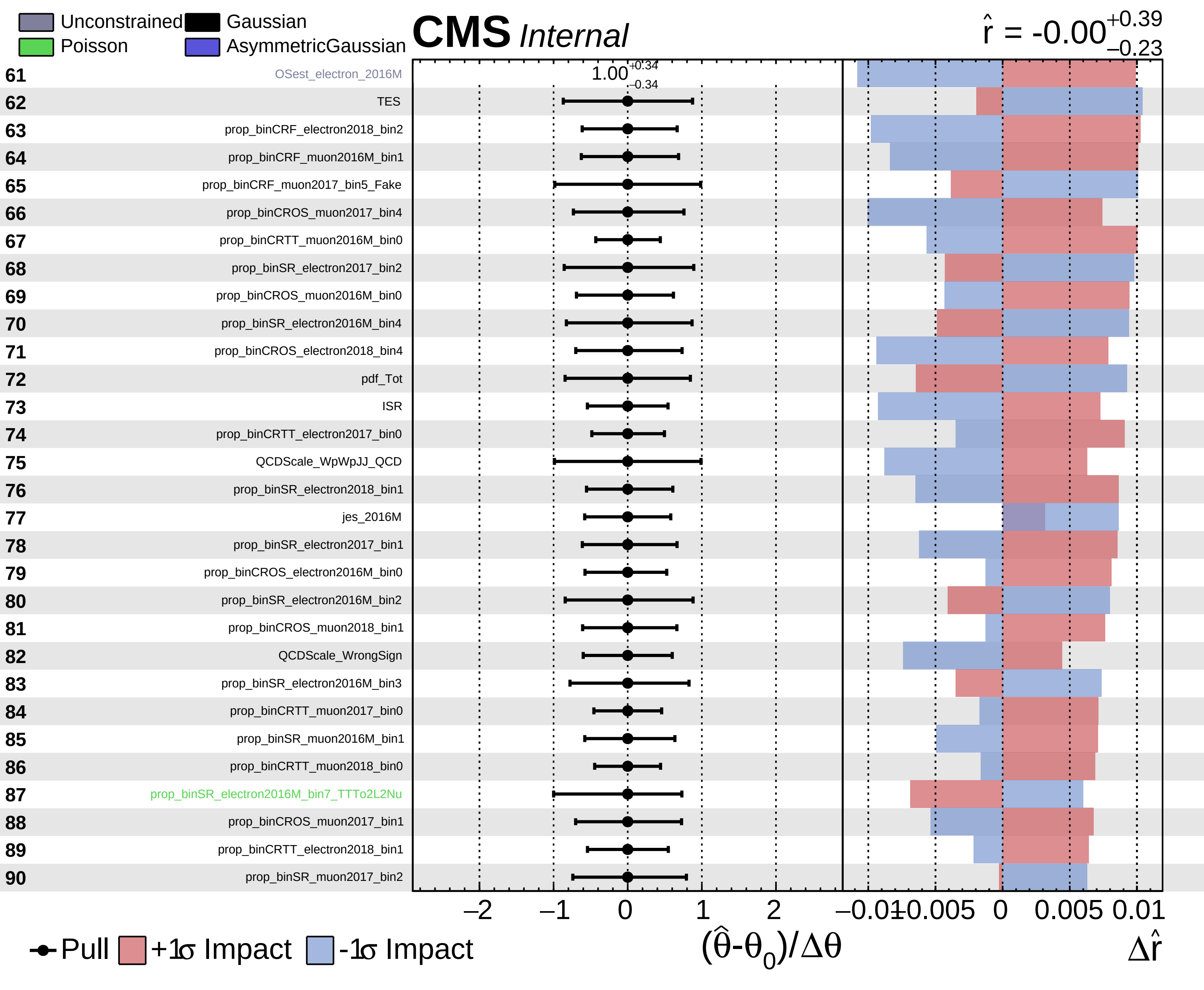


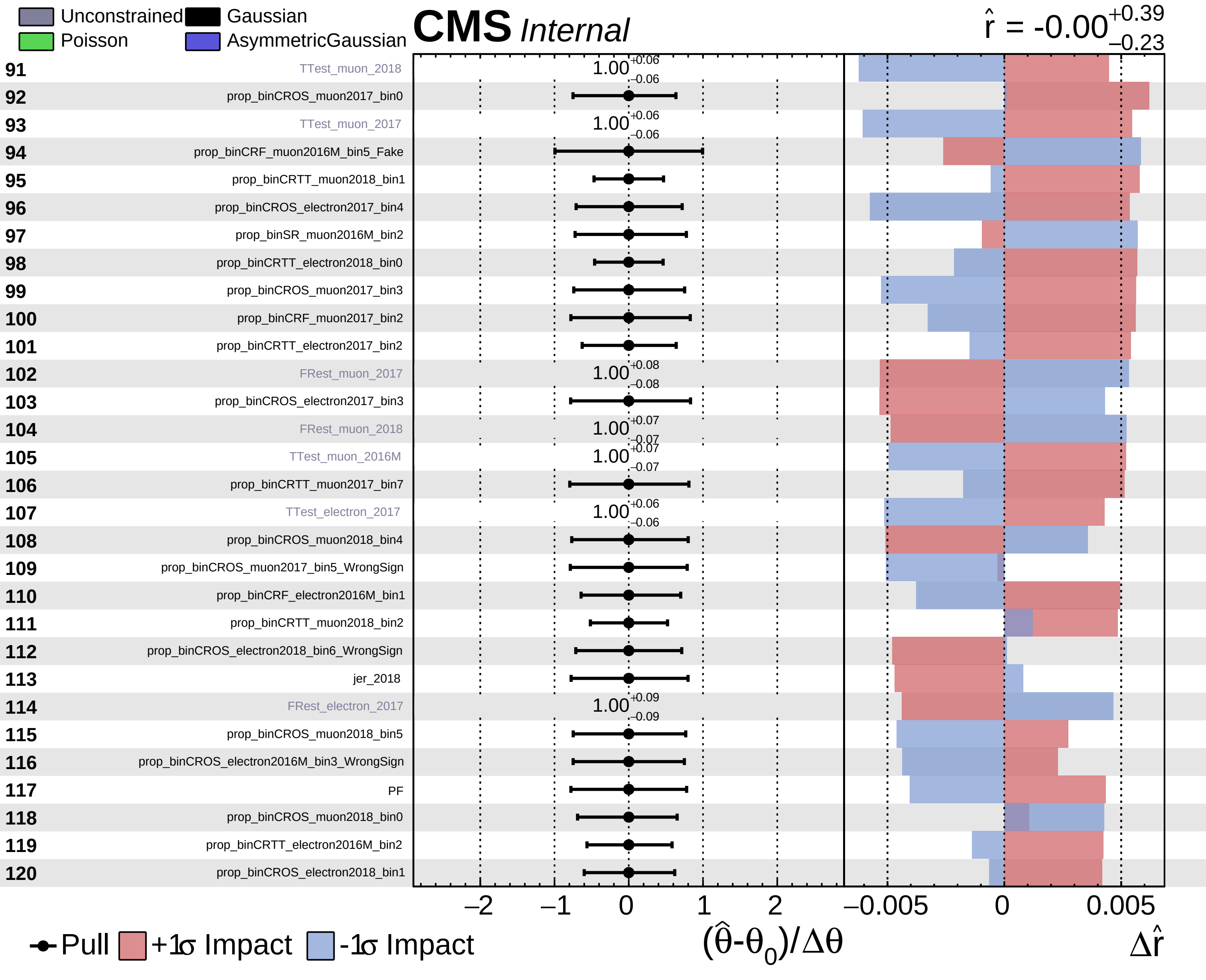
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
1 prop_binCRF_muon2018_bin0
2 prop_binCRF_muon2017_bin0
3 prop_binSR_muon2018_bin7
4 prop_binSR_muon2018_bin0
5 prop_binCRF_electron2017_bin0
6 prop_binCRF_electron2018_bin0
7 prop_binSR_muon2017_bin0
8 prop_binCRF_muon2016M_bin0
9 prop_binSR_muon2018_bin6
10 prop_binSR_muon2018_bin5
11 prop_binSR_muon2018_bin4
12 prop_binCRF_muon2018_bin1
13 prop_binSR_muon2017_bin7
14 prop_binSR_electron2018_bin6
15 prop_binSR_muon2017_bin5
16 prop_binSR_muon2016M_bin7
17 prop_binSR_electron2018_bin5
18 prop_binSR_electron2018_bin0
19 prop_binSR_muon2017_bin6
20 prop_binSR_electron2017_bin0
21 OSest_muon_2018
$1.00^{+0.22}_{-0.22}$
22 prop_binSR_electron2018_bin7
23 prop_binSR_muon2016M_bin6
24 prop_binSR_muon2016M_bin0
25 prop_binCRF_electron2016M_bin0
26 prop_binSR_muon2016M_bin5
27 OSest_muon_2017
$1.00^{+0.28}_{-0.28}$
28 prop_binSR_electron2017_bin6
29 prop_binSR_muon2018_bin1
30 OSest_electron_2017
$1.00^{+0.31}_{-0.31}$
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



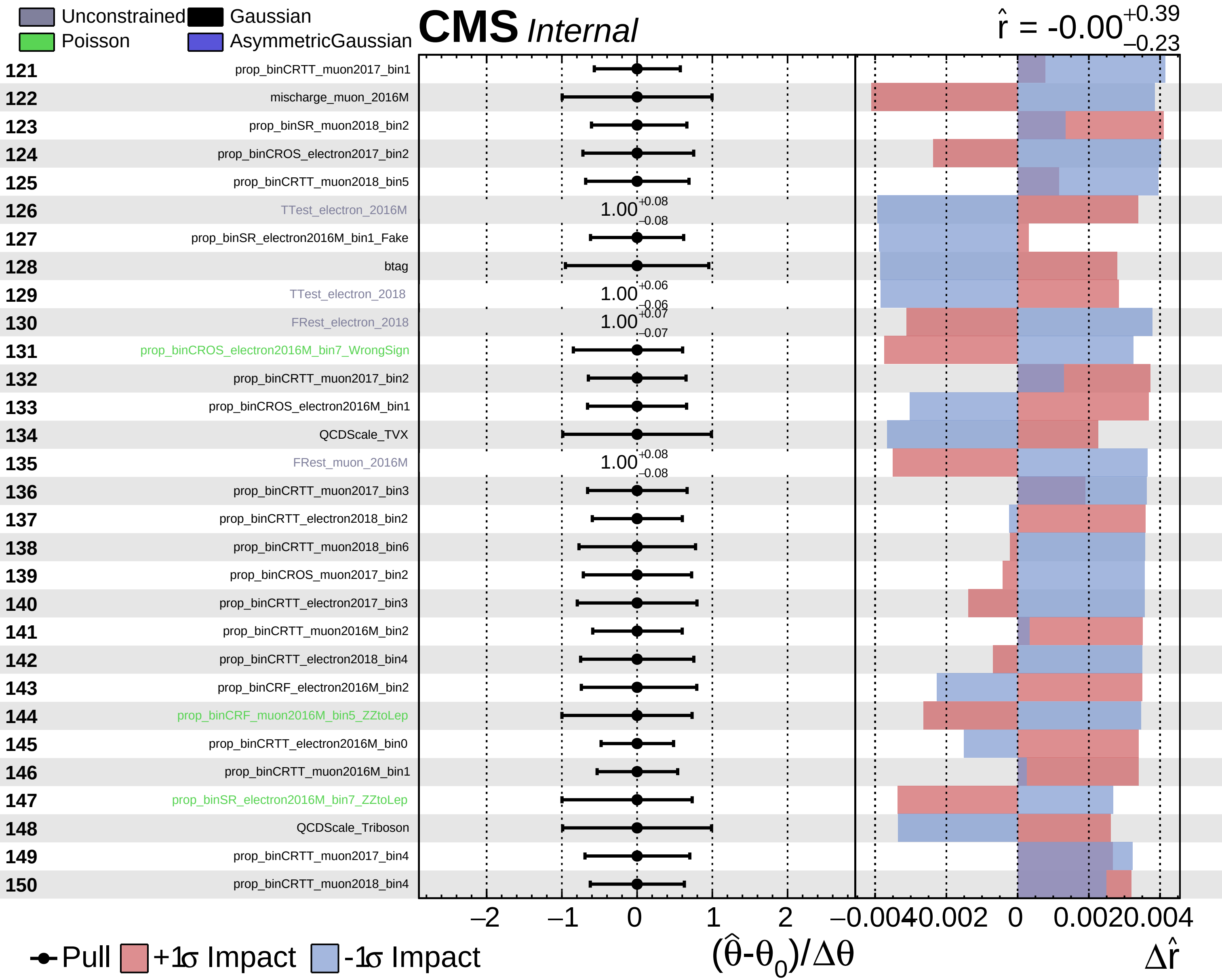
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
31 prop_binSR_electron2017_bin7
32 prop_binCROS_muon2018_bin2
33 prop_binSR_muon2017_bin4
34 prop_binSR_electron2017_bin5
35 prop_binCRF_electron2018_bin1
36 OSest_electron_2018
$1.0^{+0.4}_{-0.4}$
37 prop_binSR_muon2017_bin3
38 prop_binSR_electron2016M_bin7_Fake
39 prop_binCRF_electron2017_bin1
40 prop_binCROS_electron2017_bin0
41 jer_2016M
42 prop_binCRF_muon2017_bin1
43 prop_binSR_electron2018_bin3
44 prop_binSR_electron2018_bin4
45 prop_binSR_electron2016M_bin7_WZ
46 prop_binSR_muon2017_bin1
47 prop_binCROS_muon2016M_bin1
48 prop_binSR_muon2016M_bin4
49 prop_binSR_electron2017_bin3
50 prop_binCRF_muon2018_bin2
51 metUnclust
52 prop_binSR_electron2016M_bin5
53 prop_binCRF_muon2017_bin5_WZ
54 prop_binSR_muon2018_bin3
55 prop_binSR_muon2016M_bin3
56 prop_binSR_electron2016M_bin6
57 prop_binSR_electron2016M_bin0
58 OSest_muon_2016M
$1.00^{+0.32}_{-0.32}$
59 prop_binCROS_electron2018_bin0
60 prop_binSR_electron2017_bin4
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



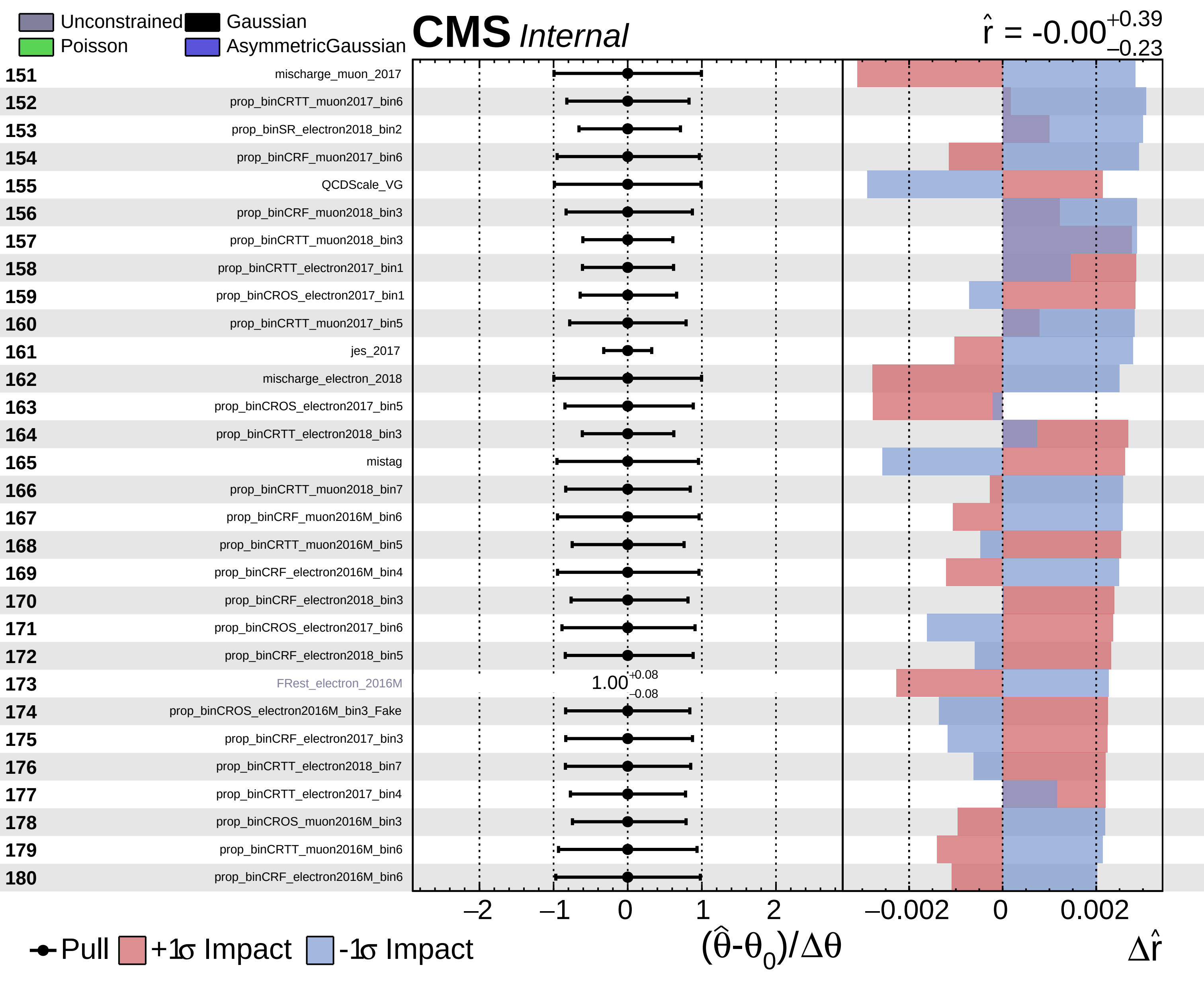
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.39 −0.23
61 OSest_electron_2016M
1.00 +0.34 −0.34
62 TES
63 prop_binCRF_electron2018_bin2
64 prop_binCRF_muon2016M_bin1
65 prop_binCRF_muon2017_bin5_Fake
66 prop_binCROS_muon2017_bin4
67 prop_binCRTT_muon2016M_bin0
68 prop_binSR_electron2017_bin2
69 prop_binCROS_muon2016M_bin0
70 prop_binSR_electron2016M_bin4
71 prop_binCROS_electron2018_bin4
72 pdf_Tot
73 ISR
74 prop_binCRTT_electron2017_bin0
75 QCDScale_WpWpJJ_QCD
76 prop_binSR_electron2018_bin1
77 jes_2016M
78 prop_binSR_electron2017_bin1
79 prop_binCROS_electron2016M_bin0
80 prop_binSR_electron2016M_bin2
81 prop_binCROS_muon2018_bin1
82 QCDScale_WrongSign
83 prop_binSR_electron2016M_bin3
84 prop_binCRTT_muon2017_bin0
85 prop_binSR_muon2016M_bin1
86 prop_binCRTT_muon2018_bin0
87 prop_binSR_electron2016M_bin7_TTTo2L2Nu
88 prop_binCROS_muon2017_bin1
89 prop_binCRTT_electron2018_bin1
90 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



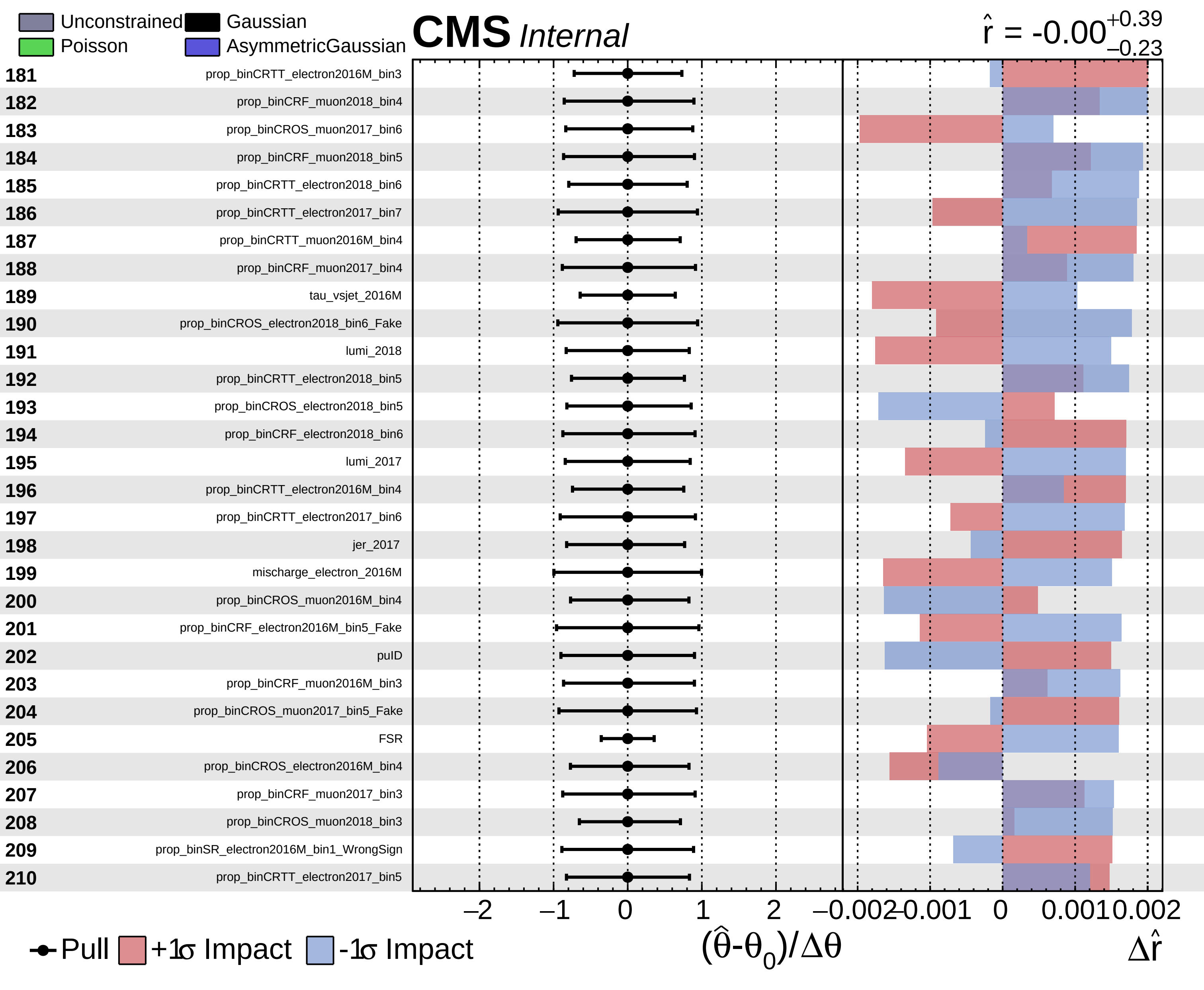
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
91 TTest_muon_2018 $1.00^{+0.06}_{-0.06}$
92 prop_binCROS_muon2017_bin0
93 TTest_muon_2017 $1.00^{+0.06}_{-0.06}$
94 prop_binCRF_muon2016M_bin5_Fake
95 prop_binCRTT_muon2018_bin1
96 prop_binCROS_electron2017_bin4
97 prop_binSR_muon2016M_bin2
98 prop_binCRTT_electron2018_bin0
99 prop_binCROS_muon2017_bin3
100 prop_binCRF_muon2017_bin2
101 prop_binCRTT_electron2017_bin2
102 FRest_muon_2017 $1.00^{+0.08}_{-0.08}$
103 prop_binCROS_electron2017_bin3
104 FRest_muon_2018 $1.00^{+0.07}_{-0.07}$
105 TTest_muon_2016M $1.00^{+0.07}_{-0.07}$
106 prop_binCRTT_muon2017_bin7
107 TTest_electron_2017 $1.00^{+0.06}_{-0.06}$
108 prop_binCROS_muon2018_bin4
109 prop_binCROS_muon2017_bin5_WrongSign
110 prop_binCRF_electron2016M_bin1
111 prop_binCRTT_muon2018_bin2
112 prop_binCROS_electron2018_bin6_WrongSign
113 jer_2018
114 FRest_electron_2017 $1.00^{+0.09}_{-0.09}$
115 prop_binCROS_muon2018_bin5
116 prop_binCROS_electron2016M_bin3_WrongSign
117 PF
118 prop_binCROS_muon2018_bin0
119 prop_binCRTT_electron2016M_bin2
120 prop_binCROS_electron2018_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



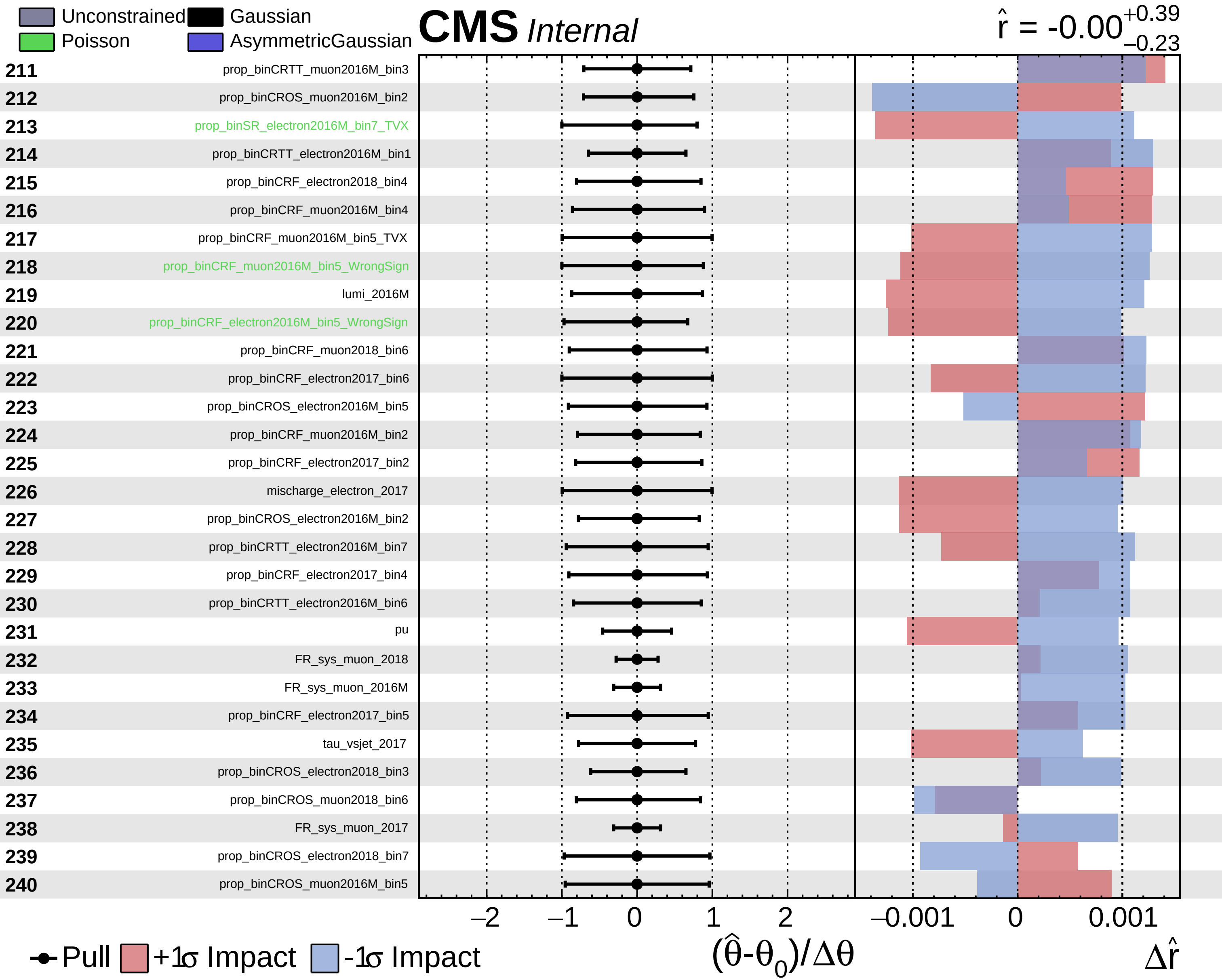
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
121 prop_binCRTT_muon2017_bin1
122 mischarge_muon_2016M
123 prop_binSR_muon2018_bin2
124 prop_binCROS_electron2017_bin2
125 prop_binCRTT_muon2018_bin5
126 TTest_electron_2016M 1.00 +0.08 −0.08
127 prop_binSR_electron2016M_bin1_Fake
128 btag
129 TTest_electron_2018 1.00 +0.06 −0.06
130 FRest_electron_2018 1.00 +0.07 −0.07
131 prop_binCROS_electron2016M_bin7_WrongSign
132 prop_binCRTT_muon2017_bin2
133 prop_binCROS_electron2016M_bin1
134 QCDScale_TVX
135 FRest_muon_2016M 1.00 +0.08 −0.08
136 prop_binCRTT_muon2017_bin3
137 prop_binCRTT_electron2018_bin2
138 prop_binCRTT_muon2018_bin6
139 prop_binCROS_muon2017_bin2
140 prop_binCRTT_electron2017_bin3
141 prop_binCRTT_muon2016M_bin2
142 prop_binCRTT_electron2018_bin4
143 prop_binCRF_electron2016M_bin2
144 prop_binCRF_muon2016M_bin5_ZZtoLep
145 prop_binCRTT_electron2016M_bin0
146 prop_binCRTT_muon2016M_bin1
147 prop_binSR_electron2016M_bin7_ZZtoLep
148 QCDScale_Triboson
149 prop_binCRTT_muon2017_bin4
150 prop_binCRTT_muon2018_bin4
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



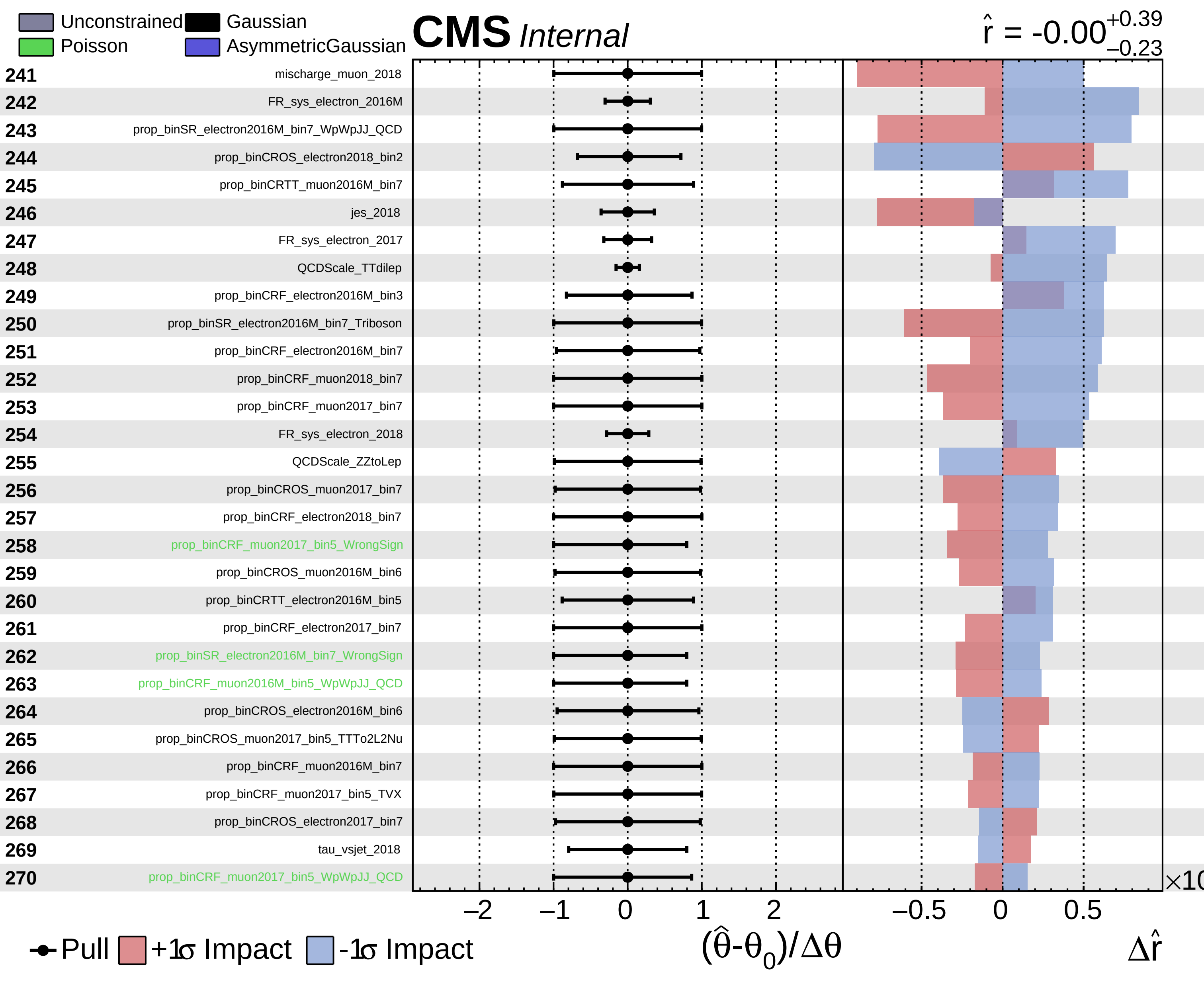
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
151 mischarge_muon_2017
152 prop_binCRTT_muon2017_bin6
153 prop_binSR_electron2018_bin2
154 prop_binCRF_muon2017_bin6
155 QCDScale_VG
156 prop_binCRF_muon2018_bin3
157 prop_binCRTT_muon2018_bin3
158 prop_binCRTT_electron2017_bin1
159 prop_binCROS_electron2017_bin1
160 prop_binCRTT_muon2017_bin5
161 jes_2017
162 mischarge_electron_2018
163 prop_binCROS_electron2017_bin5
164 prop_binCRTT_electron2018_bin3
165 mistag
166 prop_binCRTT_muon2018_bin7
167 prop_binCRF_muon2016M_bin6
168 prop_binCRTT_muon2016M_bin5
169 prop_binCRF_electron2016M_bin4
170 prop_binCRF_electron2018_bin3
171 prop_binCROS_electron2017_bin6
172 prop_binCRF_electron2018_bin5
173 FRest_electron_2016M
$1.00^{+0.08}_{-0.08}$
174 prop_binCROS_electron2016M_bin3_Fake
175 prop_binCRF_electron2017_bin3
176 prop_binCRTT_electron2018_bin7
177 prop_binCRTT_electron2017_bin4
178 prop_binCROS_muon2016M_bin3
179 prop_binCRTT_muon2016M_bin6
180 prop_binCRF_electron2016M_bin6
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

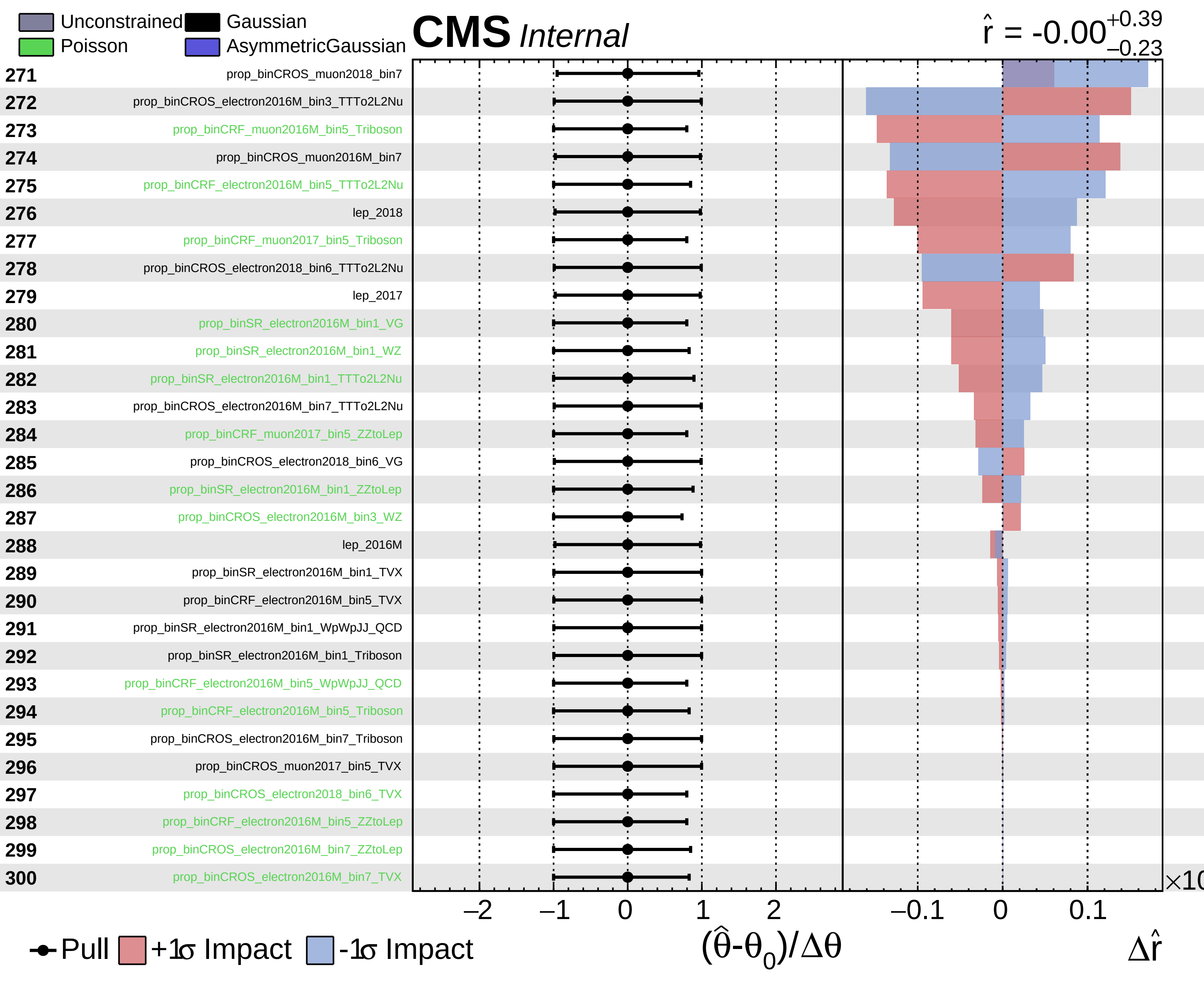


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
181 prop_binCRTT_electron2016M_bin3
182 prop_binCRF_muon2018_bin4
183 prop_binCROS_muon2017_bin6
184 prop_binCRF_muon2018_bin5
185 prop_binCRTT_electron2018_bin6
186 prop_binCRTT_electron2017_bin7
187 prop_binCRTT_muon2016M_bin4
188 prop_binCRF_muon2017_bin4
189 tau_vsjet_2016M
190 prop_binCROS_electron2018_bin6_Fake
191 lumi_2018
192 prop_binCRTT_electron2018_bin5
193 prop_binCROS_electron2018_bin5
194 prop_binCRF_electron2018_bin6
195 lumi_2017
196 prop_binCRTT_electron2016M_bin4
197 prop_binCRTT_electron2017_bin6
198 jer_2017
199 mischarge_electron_2016M
200 prop_binCROS_muon2016M_bin4
201 prop_binCRF_electron2016M_bin5_Fake
202 puID
203 prop_binCRF_muon2016M_bin3
204 prop_binCROS_muon2017_bin5_Fake
205 FSR
206 prop_binCROS_electron2016M_bin4
207 prop_binCRF_muon2017_bin3
208 prop_binCROS_muon2018_bin3
209 prop_binSR_electron2016M_bin1_WrongSign
210 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

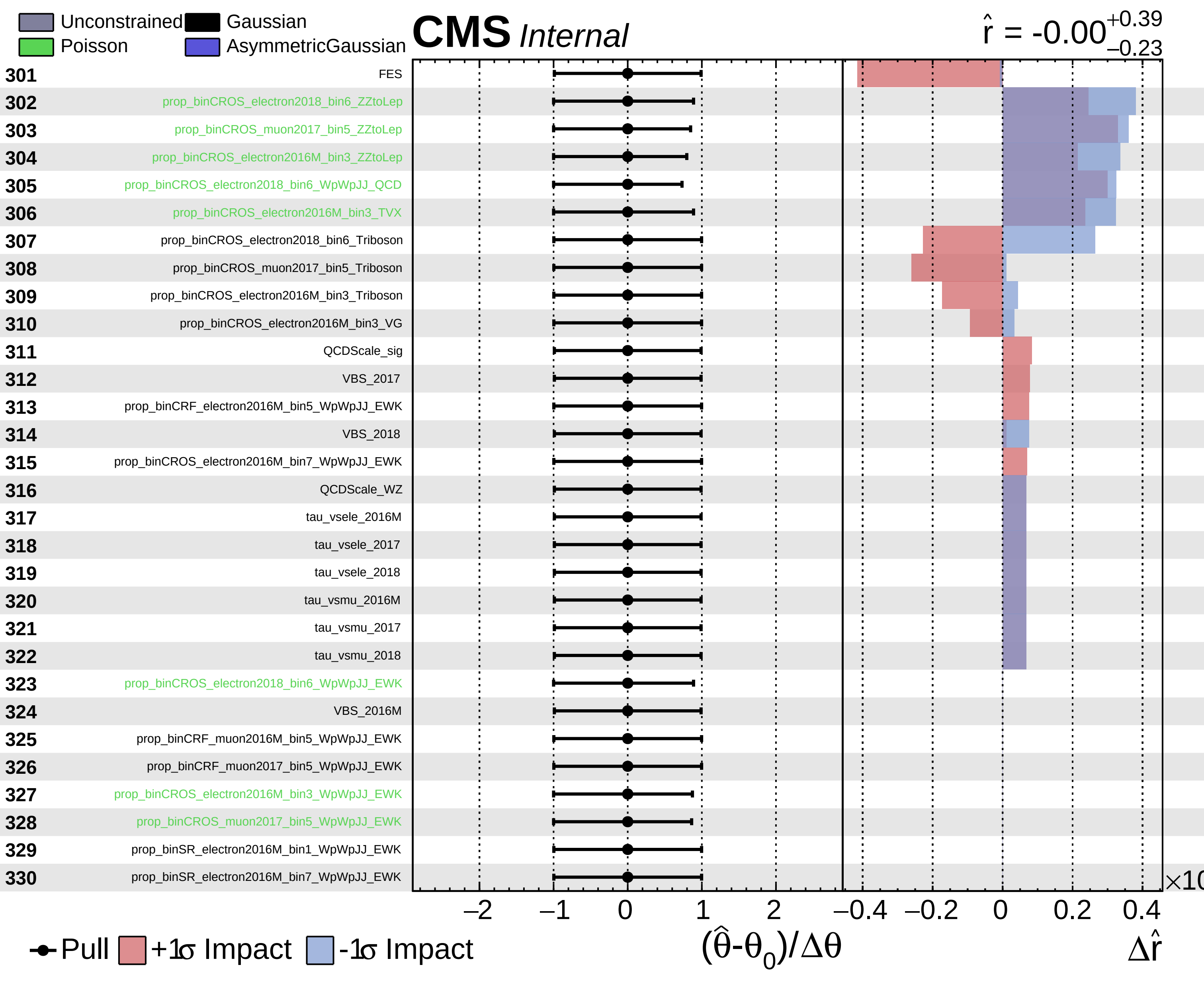


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
211 prop_binCRTT_muon2016M_bin3
212 prop_binCROS_muon2016M_bin2
213 prop_binSR_electron2016M_bin7_TVX
214 prop_binCRTT_electron2016M_bin1
215 prop_binCRF_electron2018_bin4
216 prop_binCRF_muon2016M_bin4
217 prop_binCRF_muon2016M_bin5_TVX
218 prop_binCRF_muon2016M_bin5_WrongSign
219 lumi_2016M
220 prop_binCRF_electron2016M_bin5_WrongSign
221 prop_binCRF_muon2018_bin6
222 prop_binCRF_electron2017_bin6
223 prop_binCROS_electron2016M_bin5
224 prop_binCRF_muon2016M_bin2
225 prop_binCRF_electron2017_bin2
226 mischarge_electron_2017
227 prop_binCROS_electron2016M_bin2
228 prop_binCRTT_electron2016M_bin7
229 prop_binCRF_electron2017_bin4
230 prop_binCRTT_electron2016M_bin6
231 pu
232 FR_sys_muon_2018
233 FR_sys_muon_2016M
234 prop_binCRF_electron2017_bin5
235 tau_vsjet_2017
236 prop_binCROS_electron2018_bin3
237 prop_binCROS_muon2018_bin6
238 FR_sys_muon_2017
239 prop_binCROS_electron2018_bin7
240 prop_binCROS_muon2016M_bin5
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
241 mischarge_muon_2018
242 FR_sys_electron_2016M
243 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
244 prop_binCROS_electron2018_bin2
245 prop_binCRTT_muon2016M_bin7
246 jes_2018
247 FR_sys_electron_2017
248 QCDScale_TTdilep
249 prop_binCRF_electron2016M_bin3
250 prop_binSR_electron2016M_bin7_Triboson
251 prop_binCRF_electron2016M_bin7
252 prop_binCRF_muon2018_bin7
253 prop_binCRF_muon2017_bin7
254 FR_sys_electron_2018
255 QCDScale_ZZtoLep
256 prop_binCROS_muon2017_bin7
257 prop_binCRF_electron2018_bin7
258 prop_binCRF_muon2017_bin5_WrongSign
259 prop_binCROS_muon2016M_bin6
260 prop_binCRTT_electron2016M_bin5
261 prop_binCRF_electron2017_bin7
262 prop_binSR_electron2016M_bin7_WrongSign
263 prop_binCRF_muon2016M_bin5_WpWpJJ_QCD
264 prop_binCROS_electron2016M_bin6
265 prop_binCROS_muon2017_bin5_TTTo2L2Nu
266 prop_binCRF_muon2016M_bin7
267 prop_binCRF_muon2017_bin5_TVX
268 prop_binCROS_electron2017_bin7
269 tau_vsjet_2018
270 prop_binCRF_muon2017_bin5_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.00 +0.39 −0.23
271 prop_binCROS_muon2018_bin7
272 prop_binCROS_electron2016M_bin3_TTTo2L2Nu
273 prop_binCRF_muon2016M_bin5_Triboson
274 prop_binCROS_muon2016M_bin7
275 prop_binCRF_electron2016M_bin5_TTTo2L2Nu
276 lep_2018
277 prop_binCRF_muon2017_bin5_Triboson
278 prop_binCROS_electron2018_bin6_TTTo2L2Nu
279 lep_2017
280 prop_binSR_electron2016M_bin1_VG
281 prop_binSR_electron2016M_bin1_WZ
282 prop_binSR_electron2016M_bin1_TTTo2L2Nu
283 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
284 prop_binCRF_muon2017_bin5_ZZtoLep
285 prop_binCROS_electron2018_bin6_VG
286 prop_binSR_electron2016M_bin1_ZZtoLep
287 prop_binCROS_electron2016M_bin3_WZ
288 lep_2016M
289 prop_binSR_electron2016M_bin1_TVX
290 prop_binCRF_electron2016M_bin5_TVX
291 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
292 prop_binSR_electron2016M_bin1_Triboson
293 prop_binCRF_electron2016M_bin5_WpWpJJ_QCD
294 prop_binCRF_electron2016M_bin5_Triboson
295 prop_binCROS_electron2016M_bin7_Triboson
296 prop_binCROS_muon2017_bin5_TVX
297 prop_binCROS_electron2018_bin6_TVX
298 prop_binCRF_electron2016M_bin5_ZZtoLep
299 prop_binCROS_electron2016M_bin7_ZZtoLep
300 prop_binCROS_electron2016M_bin7_TVX
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.39}_{-0.23}$
301 FES
302 prop_binCROS_electron2018_bin6_ZZtoLep
303 prop_binCROS_muon2017_bin5_ZZtoLep
304 prop_binCROS_electron2016M_bin3_ZZtoLep
305 prop_binCROS_electron2018_bin6_WpWpJJ_QCD
306 prop_binCROS_electron2016M_bin3_TVX
307 prop_binCROS_electron2018_bin6_Triboson
308 prop_binCROS_muon2017_bin5_Triboson
309 prop_binCROS_electron2016M_bin3_Triboson
310 prop_binCROS_electron2016M_bin3_VG
311 QCDScale_sig
312 VBS_2017
313 prop_binCRF_electron2016M_bin5_WpWpJJ_EWK
314 VBS_2018
315 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
316 QCDScale_WZ
317 tau_vsele_2016M
318 tau_vsele_2017
319 tau_vsele_2018
320 tau_vsmu_2016M
321 tau_vsmu_2017
322 tau_vsmu_2018
323 prop_binCROS_electron2018_bin6_WpWpJJ_EWK
324 VBS_2016M
325 prop_binCRF_muon2016M_bin5_WpWpJJ_EWK
326 prop_binCRF_muon2017_bin5_WpWpJJ_EWK
327 prop_binCROS_electron2016M_bin3_WpWpJJ_EWK
328 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
329 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
330 prop_binSR_electron2016M_bin7_WpWpJJ_EWK
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$